Impact of AI Tools on Learning Behavior of College Students

Abstract—Artificial Intelligence (AI) tools such as ChatGPT, Gemini, DeepSeek and Copilot are increasingly being adopted by college students for academic purposes. These tools provide easy explanation in assignments, support coding tasks and research activities. With the rapid use of AI-driven platforms, traditional learning methods are undergoing some major changes, that is, students are no longer dependent on textbooks and classroom lectures but are actively integrating AI assistants into their daily study routines. Despite the huge productivity gains, critical concerns have been recognized regarding over-dependence and the use of independent problem-solving capabilities. In this study survey-based research on the impact of AI tools on the learning behavior, study habits and academic performance of undergraduate college students is done. A total of approx. ~a hundred students from different academic fields participated in the survey. The data was analyzed using statistics and percentage analysis. On finding that while AI tools measurably improve learning efficiency and time management, they simultaneously introduce risks of dependency and critical thinking. The study recommends a balanced framework for AI integration in higher education, one that preserves the foundational essential skills for long-term academic and professional success**.**

Keywords— Artificial Intelligence, AI Tools, Learning behavior, College Students, Educational Technology, Higher Education, Critical Thinking

1 Introduction

The formation of large language models (LLMs) and AI powered assistants marks one of the most significant importance educational technologies. Tools such as OpenAI's ChatGPT, Google's Gemini and DeepSeek research applications are making more into mainstream academic usage. Within a matter of months following the public release of ChatGPT in November 2022, surveys across multiple universities reported that a substantial proportion of students had already incorporated the tool into their coursework [1, 2].

Earlier, students relied on knowledge sources like textbooks, lecture notes, library and teacher consultations, each having levels of effort and critical engagement.

This shift in behavior raises questions for educators and researchers that

- Does AI-mediated learning enhance genuine comprehension, or does it primarily optimise surface-level task completion?
- Does it democratise access to knowledge, or does it widen skill-development gaps between students who use it thoughtfully and those who rely on it passively?

Several prior studies have begun to address these questions. Zawacki-Richter et al. [3] conducted a systematic review of AI applications in higher education and identified personalized learning and intelligent tutoring systems as the most common use cases. Luckin et al. [4] argued that AI holds the potential to transform education by providing adaptive, data-driven feedback, but warned that the pedagogical framing of AI tools is crucial. Holmes et al. [5] further highlighted the risks of uncritical AI adoption, including threats to assessment integrity and the risk of epistemic dependence.

Despite this few studies have focused specifically on undergraduate students in emerging economies, where access to AI tools may interact differently with existing educational infrastructure, language barriers, and assessment cultures. This study addresses that gap by focusing on students at – a context where AI tool adoption among students has accelerated markedly since 2023.

2 Objectives

The primary objectives are:

- This study is done to understand how frequently students across different situations are using AI tools in their day-to-day academic life.
- To look into how AI tools are affecting the way students study, manage their time, and approach learning in general.
- To find out whether using AI tools actually helps students understand their subjects better, perform well and prepare more effectively for exams.
- To highlight the problems that come with AI tool usage, such as becoming too dependent on them and losing the ability to think and solve problems independently.
- To suggest practical ways in which AI tools can be used responsibly.

3 Related Work and Theoretical Framework

To understand how and why students adopt AI tools in their studies, the study helps to look at two important theories that researchers have used in this area. The first is the Technology Acceptance Model or TAM, was introduced by Davis, which basically says that people are more likely to use a new technology if they find it useful and easy to use. The second is Vygotsky's idea of the Zone of Proximal Development, which says that students learn best when they get just enough support to push past what they can do on their own, and that this support should slowly be reduced as the student becomes more capable.

Zawacki-Richter et al. [3] carried out a large review of 146 published articles on AI in higher education and found that most of the existing research focused on things like predicting student performance or identifying students who might be struggling. Very few studies actually looked at whether AI was helping students think more deeply or engage meaningfully with their learning. This shows that there is still a big gap in research when it comes to understanding the quality of learning that happens when students use AI tools.

Kasneci et al. [8] looked specifically at ChatGPT and similar AI language tools in educational settings and found both positives and negatives. On the positive side, these tools can offer personalized feedback and support students who have learning difficulties. On the negative side, they can sometimes produce information that sounds convincing but is actually wrong, they can make it easier for students to be academically dishonest, and they might discourage students from putting in the mental effort needed for real understanding. The researchers stressed the need for clear, evidence-based guidelines on how AI should be used in academic settings.

Closer to home, Sharma and Kumar [9] studied students in Indian engineering programmes and found that they mostly used AI tools for writing and fixing code, and occasionally for getting explanations of difficult concepts. This is an important finding because it suggests that how students use AI can vary quite a lot depending on what subject they are studying, and that patterns seen in Western universities may not always apply to the Indian context.

Taken together, the existing literature shows that AI tools are a double-edged sword. The very things that make them so helpful the fact that they are always available, can answer almost anything, and respond instantly are also what make them easy to misuse.

4 Methodology

This study adopts a quantitative, survey-based research design. Survey methods are well-suited to capturing self-reported attitudes, behaviors and perceptions at scale, making them appropriate for exploring the breadth of AI tool usage patterns across a heterogeneous student population [10].

4.1 Research Design and Participants

A survey among students of , was conducted of the academic year 2023–2024. Participants were given a google form via linked shared among class groups from multiple departments like Computer Science, Mechanical Engineering, Business Administration and etc. A total of approx. 100 valid responses were collected. Participation was entirely voluntary and consent was obtained from all participants prior to data collection.

4.2 Instrument Development

A questionnaire was formed with our 4 team members. The questionnaire comprised four sections
(i) Information regarding year of study and department
(ii) AI tool usage patterns, frequency of use and tools usage
(iii) Any effects on learning habits and academic performance (rated on a five-point Likert scale ranging from Strongly Agree to Strongly Disagree)
(iv) open-ended items capturing self-reported benefits and challenges. The Cronbach's alpha reliability coefficient for the Likert-scale section was calculated at 0.83, indicating good internal consistency.

4.3 Data Collection

We collected the data into google form and then filtered all the responses. There around 100 students who filled the google form properly and from that data we checked it and we collected the data. Approx. 62 were male and 38 were females, 45 students from first and second year, 55 were from third year and fourth year. This give us a mix of responses from different levels and streams.

4.4 Data Analysis

Data from the Likert-scale items were analysed, including percentage distributions. Responses were into these categories – 'Strongly Agree/Agree', 'Neutral' and 'Disagree/Strongly Disagree' to get a clear response. All analyses were performed using Microsoft Excel.

5 Results and Findings

The results are presented in three thematic clusters:
(i) patterns of AI tool adoption
(ii) perceived benefits of AI usage
(iii) perceived challenges and risks.

5.1 Patterns of AI Tool Adoption

A majority of students were using AI tools at least several times per week for academic purposes, with 34% reporting daily usage. The use of Ai Tools by the students of Computer Science reported the highest percentage of use and from Business Administration reported the highest rely on AI for essay writing and report generation. The most commonly purposes for AI use were in assignment completion, understanding difficult concepts and exam preparation.

5.2 Perceived Benefits

The majority of respondents reported positive effects of AI tool usage on their learning experience. 78% agreed or strongly agreed that AI tools save time on assignments, 80% agreed that AI helps them understand difficult topics more easily and 73% reported improved exam preparation. Several responses the personalized nature of AI explanations were a key benefit to the students, noticing that AI tools can explain to their current level of understanding that textbooks cannot.

5.3 Perceived Challenges and Risks

Despite these benefits, a substantial proportion of respondents acknowledged concerning patterns of AI dependence. Sixty-seven percent agreed or strongly agreed that over-dependence on AI tools is a genuine concern, and 58% reported that they sometimes submit AI-generated content without thoroughly reviewing or understanding it. Forty-two percent acknowledged that their use of AI has reduced their tendency to engage in independent problem-solving. Thematic analysis of open-ended responses revealed three recurring concerns:
(i) the risk of receiving incorrect or incomplete information from AI tools, particularly for specialized academic topics
(ii) uncertainty about the ethical boundaries of AI use in assignments
(iii) a perceived reduction in the depth of engagement with course material.

5.4 Summary Table

TABLE 1 PRESENTS A SUMMARY OF THE SURVEY RESPONSES ACROSS KEY ATTITUDINAL ITEMS

Table 1. Survey Responses on AI Tool Usage and Perceived Impact (N = 100).

Statement	Strongly Agree	Agree	Neutral	Disagree
Regular use of AI tools for studies	48%	30%	14%	4%
AI helps understand complex topics	42%	38%	12%	8%
AI saves time on assignments	55%	28%	10%	7%
AI improves exam preparation	38%	35%	17%	10%
Over-dependence on AI is a concern	35%	32%	20%	13%
AI reduces independent thinking	30%	28%	25%	17%
Willingness to limit AI use if advised	28%	34%	24%	14%

** Percentages may not sum to 100% due to rounding.*

6 Discussion

The result of this study shows a mixed responses of how AI tools are being used by students. On the one side, students find AI tools helpful. They save time, make difficult topics easier to understand. This matches what other researchers have found in studies done in different parts of the world, where students reported similar benefits from using AI in their studies.
However, on the other side to these findings, the fact that 58% of students sometimes submit AI-generated content without properly reading or checking it. Students are not really learning they are just copying an output without engaging to learn it by themselves. This is a problem because real learning happens when a student struggles with something, thinks it thoroughly and arrives with an understanding on their own. If AI is always doing that thinking for them, they may never actually develop those skills. We also noticed that students from different fields use AI in different ways. Computer Science students used AI the most, mainly for help with coding.

This makes sense because there are many AI tools specifically built for programming. But this also raises a concern if students rely too heavily on AI to write their code, they may never properly learn how to think through a problem logically or figure out errors on their own. These are skills that are essential for anyone who wants to work as a software engineer.

Another important point is students own written responses were confusion for what is allowed. Many students were not sure whether what the work they were doing was acceptable or not. This is because most colleges and universities have not yet set rules about AI use. This lacks in clarity itself and problems without proper guidelines, students might accidentally cross a line and do something their institution considers cheating or on the other hand, they might using AI even in situations where it would be completely fine and helpful.

Finally, it is important to be honest about the limitations of this study. All responses came from students at a single college, so the findings may not represent the experiences of all undergraduate students across India. Since students filled out the questionnaire themselves, there is always a chance that some answers were not entirely accurate. Also, because the data was collected at one point in time, we cannot say for certain what is causing what. Future research should follow students over a longer period, include multiple colleges, and use more objective ways of measuring academic performance to get a clearer and more reliable picture.

7 Conclusion

This study, the impact of AI tools on the learning behavior of college students through a survey of 100 participants across multiple streams. The findings demonstrate that AI tools have become deeply embedded in students academic practices as well as offering genuine benefits in terms of efficiency, conceptual accessibility and exam preparation. At the same time the data reveal concerns about dependence, more consumption and uncertainty that, if unaddressed, may undermine the long-term intellectual development of students.

The study concludes that AI tools in education are most beneficial when they are used as active scaffolding that gradually transfers cognitive responsibility back to the learner – a process that must be intentionally designed into curricula and assessment practices. Universities should develop clear, evidence-based guidelines for AI use, train faculty to incorporate AI literacy into their teaching, and create assessment environments that reward the uniquely human cognitive skills critical analysis, synthesis, and creative problem-solving that AI tools cannot replicate.

8 Future Scope and Recommendations

The increasing integration of Artificial Intelligence tools in higher education indicates that their influence on learning behavior will continue to grow in the coming years. Future research should investigate the long-term academic and cognitive effects of AI-assisted learning among students from diverse educational backgrounds. Studies involving multiple universities and larger datasets may provide a more comprehensive understanding of how AI tools influence critical thinking, creativity, problem-solving ability, and academic integrity.

Another important direction for future work is the development of institutional guidelines and ethical frameworks regarding the responsible use of AI in education. Universities should focus on AI literacy programs that teach students how to use AI tools as learning assistants rather than as replacements for independent thinking. Faculty members should also receive training on redesigning assessments and assignments in ways that encourage analytical reasoning and originality.

Further studies may also compare the usage patterns of AI tools across different disciplines such as engineering, business, humanities, and medical education. Understanding discipline-specific differences can help educators design more effective teaching strategies and balanced AI integration policies. Additionally, future work can explore the psychological impact of excessive AI dependence on student confidence, motivation, and self-learning behavior.

9 Ethical Considerations

This study ensured voluntary participation and maintained the anonymity of all participants. No personally identifiable information was collected during the survey process. The collected responses were used strictly for academic research purposes. Participants were informed about the purpose of the study before submitting their responses. The research emphasizes that AI tools should support learning in an ethical and transparent manner while preserving academic honesty and independent thinking.

10 Educational Implications

The findings of this study highlight the urgent need for higher educational institutions to redesign learning environments in response to the rapid adoption of AI tools by students. Traditional methods of evaluation that depend heavily on take-home assignments may no longer accurately measure student understanding. Institutions should increasingly adopt project-based assessments, oral evaluations, case-study discussions, and supervised practical work to encourage authentic learning.

11 Recommendations for Responsible AI Usage

Students should be encouraged to use AI tools primarily for concept clarification, brainstorming, and academic support rather than direct answer generation. Teachers can guide students on verifying AI-generated responses through textbooks, research papers, and classroom discussions. Responsible AI usage policies can help create a balanced educational ecosystem where technology supports learning without replacing critical thinking and originality. Institutions should also conduct awareness programs about ethical AI practices and academic integrity.

Overall, the integration of AI into higher education presents both opportunities and challenges. While AI tools improve efficiency and accessibility of learning resources, their long-term educational impact depends on how responsibly they are used. A balanced approach that combines technological innovation with strong pedagogical practices will be essential for ensuring meaningful student learning outcomes in the future.

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